

1. (9 pts) Give examples for each of the following:

- a) An abelian group that is not cyclic
- b) An abelian subgroup of a nonabelian group.
- c) Three elements in the group $GL(2, \mathbb{R})$

2. (9 pts) Refer to the Cayley table to answer the questions about the group G .

a) (2 pts) What is $6*5$ in G ?

b) (3 pts) List the elements in the centralizer of 5.

c) (4 pts) Find 2 subgroups of order 4 in G

	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	3	4	1	8	5	6	7
3	3	4	1	2	7	8	5	6
4	4	1	2	3	6	7	8	5
5	5	6	7	8	3	4	1	2
6	6	7	8	5	2	3	4	1
7	7	8	5	6	1	2	3	4
8	8	5	6	7	4	1	2	3

3. (9 pts) List all the elements in Z_{10} and in $U(10)$. Is $U(10)$ a subgroup of Z_{10} ? Explain your answer.

4. (9 pts) Consider the group Z_{40} .

- a) Find a subgroup of order 10.
- b) List all the elements that have order 10.
- c) List all possible orders for elements in Z_{40} .

5. (6 pts) Assume G is a finite cyclic group containing elements x , y , and z .

- a) If $x^{10} = e$ and $x^{21} = e$, what can you say about x ?
- b) If $|y| = 10$ and $|z| = 21$, what can you say about $|G|$?

6. (4 pts) Draw the subgroup lattice for $\mathbb{Z}_{40}$.

7. (6 pts) Let a be a group element such that $|a| = 48$.

- a) Find a divisor k of 48 such that $|\langle a^k \rangle| = 6$.
- b) Find a divisor n of 48 such that $\langle a^{10} \rangle = \langle a^n \rangle$

8. (8 pts) True or False explain your answer.
- a) If there exist integers $x, y, s,$ and t such that $s*x + t*y = 10$, then $\gcd(x,y) = 10$.
 - b) If $Z(G) = G$ then G must be abelian. (Recall that $Z(G)$ is the center of a group.)
9. (10 pts) Let a and b be integers with $\gcd(a,b) = d$. Prove $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$
10. (10 pts) Prove that the identity in any group is unique.
11. (10 pts) Let G be an abelian group. Let H be the set of all elements of G that satisfy the equation $x^3 = e$. Prove that H is a subgroup of G .
12. (10 pts) List all the elements of $U(25)$. Prove that $U(25)$ is cyclic and find all of its generators.